

Books-59 — The three taxes, and the email address that broke your deploy

For Michael Lyman · Greenway Marijuana · Port Orchard, WA

The short version

You gave me five things to do. Here is where each one stands.

- 1. Squash merge everything so you can inspect it.** Done and pushed. All 44 commits from books-55 through books-58 are now a single commit on `main`. I proved it carried faithfully by comparing tree hashes rather than eyeballing a file list — the squashed tree and the branch tip are the same hash, `29d6c82d`, which means not one byte was dropped or added. The full battery then ran `on main` and passed: 464 files, 11,338 tests.
 - 2. The email address that broke your deploy.** You were right, and the cause was worse than a bad setting: it was me breaking a standing rule you already had. Fixed at the root, and the deploy is green again. Details below, because you should know how it happened.
 - 3. The DOR form is cancelled.** Understood — you enter total sales and it tells you what you owe, so there is nothing to teach. I will strip it out of the roadmap so it stops promising you a page you do not want.
 - 4. B&O accounted for as its own tax type.** Done, and your July return now lives in the system as permanent test evidence. Every line of it recomputes to the cent. There is one genuine finding you need to weigh in on, in the section called "*Expense and not a liability*" — *you are right, and here is the nuance*.
 - 5. Lessons for the untaught federal boxes, then the next physical form.** Not started yet. This report is the pause before that work.
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The email address: I broke a rule you already had

Your repository is private now, and Vercel refuses to build when the commit carries an address that does not look like a person. The address on your last commit was `root@172.22.219.243` — a container IP. That is what stopped the deploy.

What makes this worth a full explanation is that **you already had a rule about this, and I had already been warned by the same failure once before**. Rule 20 in the standing rules says to commit as `Greenway Dev <dev@greenwaymarijuana.com>`. Rule 20a, added back on books-12, says specifically that *a squash merge throws the author away* — it was written after `gh pr merge --squash` did exactly this.

I then did it again by a different door. Rule 20a warned about `gh pr merge --squash`; I used `git merge --squash` followed by `git commit`. Same operation, different spelling, and I did not put the email flag on that particular command. This sandbox had no email configured at all, so git quietly invented one from the container's IP address rather than stopping to ask.

The real fault is in how the rule was written. It depended on me remembering a flag on every single commit, which is the kind of guard that works until the one time it does not. **A wrong default is more dangerous than no default**, because it produces a plausible-looking commit instead of an error you would notice.

What I changed so it cannot recur:

The identity is now set in git's own configuration — both for this clone and globally, so a fresh clone later in the same session inherits it too. Git can no longer guess, because there is nothing left to guess.

I also changed how it gets verified. Checking locally was never enough; the check that actually proves anything is asking GitHub what it stored:

```
gh api repos/mblyman89/GREENWAY-WEBSITE/commits/main --jq '.commit.author.email'
```

That now returns `dev@greenwaymarijuana.com`.

Repairing the commit that was already pushed. Your `main` branch is not protected (I checked — GitHub returns "Upgrade to GitHub Pro" for the protection API, so no protection exists), the bad commit was the very tip, and nobody had pulled it in the

few minutes it existed. That made it safe to relabel. Before force-pushing I compared the tree hash before and after the fix: both `29d6c82d` , unchanged. Only the author line moved; not a line of code was touched.

Your redeploy is triggered and green. The corrected push fired a fresh CI run which completed successfully.

One thing I did not do. Your repository's history contains a great many older commits with bad addresses — 989 from the bot account, and about forty with container IPs, going back long before this session. I left them alone. Rewriting shared history to tidy up cosmetics would be a far bigger risk than the untidiness, and only the tip actually matters to Vercel.

Your July return is now evidence, not a note

You said: *"my uploaded JULY.pdf in the workspace is the actual return i filed with dor, so it is authoritative."*

You are right that it is authoritative, and that is precisely why I checked it instead of copying it. An authority transcribed wrongly stops being an authority. So before a single figure was used, every line was recomputed from its own base and rate:

Line	Base	Rate	Computed	You filed	
B&O Retailing	168,465.17	0.004710	793.47	793.47	match
B&O Service & Other (ATM)	4,355.00	0.015000	65.33	65.33	match
State Retail Sales	168,465.17	0.065000	10,950.24	10,950.24	match
Local — Port Orchard 1802	168,465.17	0.028000	4,717.02	4,717.02	match
B&O subtotal			858.80	858.80	match
Grand total			16,526.06	16,526.06	match

Six for six, to the cent. **Your return foots perfectly.**

I also found a second copy of the same return already in the workspace (`July 2026 Department of Revenue.pdf`), diffed the two, and confirmed they are the same document. And I re-extracted from the PDF rather than trusting a text file that was already sitting there from an earlier session.

The return is now mirrored in the repository at `docs/authorities/state-wa/dor-combined-excise-return-july-2026.txt` , byte-for-byte identical to what comes out of the PDF, alongside the IRS and FASB sources. Its confirmation number (0-053-958-352) is pinned in the tests, so any figure can always be traced back to one specific real filing.

The thing I did not expect: your B&O rate does not fit

This is the most important technical finding of the slice, and it is a good example of why "never assume" earns its keep.

Every tax rate in your system is stored as an integer number of **basis points** — hundredths of a percent. That worked perfectly while sales tax was the only rate involved. 6.5% is exactly 650 basis points. 2.8% is exactly 280. Clean integers, no drift, no rounding.

Your B&O Retailing rate is **0.004710**. In basis points that is **47.10** — **not a whole number**. There is no way to store it in the existing unit without changing the rate, and changing the rate means disagreeing with a return you already filed with the state.

Here is what that would have cost, measured on your actual July numbers:

What we store	July B&O retailing	Versus your filing
0.004710 — the real rate	793.47	matches exactly
rounded down to 47 bps	791.79	understates by \$1.68/month
rounded up to 48 bps	808.63	overstates by \$15.16/month

A dollar sixty-eight a month is about twenty dollars a year. Small in dollars, total in meaning: your books would permanently disagree with a filed government return, and every reconciliation from then on would be chasing a difference the unit itself created. **Rounding a rate is not a rounding error. It is storing the wrong rate.**

So B&O rates are stored in **millionths** — the rate times one million. `0.004710` becomes `4710` , `0.015000` becomes `15000` . Both

exact whole numbers. Basis points convert into the same unit without moving at all (650 bps = 65,000 millionths), so **nothing that already worked changes its answer**. Everything stays an integer; only the fineness changed.

I also built the rates so they can be changed. You said it yourself — *"im sure it updates however often as it does"* — and your service rate is conditional on prior-year receipts staying under \$1,000,000, a threshold you could cross. The rates are settable defaults, never hardcoded inside a calculation where a future change could not reach them.

"Expense and not a liability" — you are right, and here is the nuance

You said: *"its important that the books account for b&o as it is an expense and not a liability."*

Your instinct is exactly right, and it is the single most important thing about B&O. Let me lay out why, and then flag one nuance I need your call on rather than deciding for you.

Why B&O is different from your other two taxes. Retail sales tax and the 37% cannabis excise are **trust money**. You collect them from the customer on the state's behalf. They were never yours — you are holding someone else's cash on its way to Olympia. So they hit a liability on the way in, clear it on the way out, and **never touch your profit and loss at all**.

B&O is not that. Nobody hands you B&O at the register. It is a tax **on Greenway**, measured by your gross receipts, and it comes out of your own pocket. **That makes it a genuine business expense that genuinely reduces your profit** — which is your point, and it is correct.

The nuance. Your chart of accounts (migration 0173, written long before this conversation) already carries *both* 75040 B&O Tax Expense **and** 32200 B&O Tax Payable . That looks like it contradicts you. It does not — the two accounts answer different questions.

The expense records **what it cost you**. The payable records **that you have not paid it yet**. Your July sales happened in July; the money left your bank on 25 August. Accrual books put the expense in July, where the sales were, and park the amount owed in the payable until the cash actually moves:

When the month closes:					
DEBIT	75040	B&O Tax Expense	858.80	<-	the expense you mean
CREDIT	32200	B&O Tax Payable	858.80	<-	just "not paid yet"

When you pay DOR:				
DEBIT	32200	B&O Tax Payable	858.80	
CREDIT	10100	the bank	858.80	

The liability is temporary plumbing that nets to zero. The expense is the economic fact, and it stays. **If B&O were booked the way sales tax is booked — straight to a liability with no expense side — your profit would be overstated by the full B&O amount every single month, forever.** That is the error you were guarding against, and the system now actively refuses it.

I have built it to keep both sides, because dropping the payable would have meant guessing at what you meant instead of reading what you said. **If you want B&O expensed directly when paid, with no payable at all, say so and I will change it.** That is a legitimate choice on a smaller set of books; it just is not the standard accrual treatment, and it is your call, not mine.

One warning attached to this. Migration 0173 flags account 75040 as `nondeductible_280e` . Being an expense in your books does **not** make B&O deductible on your federal return — §280E still disallows it for a cannabis retailer. It is a real book expense and a disallowed tax deduction at the same time. That difference belongs in the book-to-tax bridge, and this slice does not touch your return.

The real gap was not what either of us thought

Before writing anything, I measured what already existed. The result was not what I expected:

Tax type	Was it computed?	Did anything post it to the books?
State sales (6.5%)	Yes — already split out	via the trust liability
Local sales (2.8%)	Yes — already split out	via the trust liability
B&O	No . Nothing anywhere	No

Two of your three types were already handled. `wa-tax.ts` has been computing state and local separately for a long time, and your tax report page already prints both with their rates.

B&O was the hole — and a strange one. The accounts `75040` and `32200` have existed since migration 0173, complete with careful descriptions explaining that B&O "*has an expense side — unlike excise and sales tax, which are trust money and never touch the P&L.*" Someone understood this perfectly and wrote it down. Then **nothing was ever built to use it**. Searching the entire codebase for either account number returns nothing outside the chart of accounts itself, and there was no B&O rate anywhere in the system at all.

So the chart of accounts had the right idea sitting unused for months. That is now connected.

Your ATM answer, built in

You said: "*the surcharge only is what i enter into the dor portal when paying sales taxes. the atm service provider deposits the surcharge separate so it will be very simple to track.*"

That matches your July return exactly — the "Service and Other Activities" line is \$4,355.00, which is surcharge income, not withdrawal volume. Built to that shape:

- The surcharge is its own B&O classification at its own rate (1.5%), **not** the retail rate. This matters more than it looks: the service rate is over three times the retailing rate, so confusing them is a real misstatement, not a rounding matter. There is a test that fails if anyone ever taxes the ATM at the retail rate.
- The surcharge is tagged to the **ATM entity**, retail sales to **Greenway**, keeping your four-entity separation intact.
- **Cash withdrawn is never treated as revenue**. It is vault money moving. Taxing withdrawal volume as gross receipts would be a catastrophic overstatement, so the engine only ever sees surcharge income.
- A month with no surcharge produces no ATM line at all, because DOR does not print one.

Your chart of accounts already has `51000 ATM Surcharge Income` tagged to the ATM entity — and, like the B&O accounts, **nothing posts to it yet**. Worth flagging for when we wire the ATM feed.

Proving the tests can actually fail

A test that cannot fail proves nothing. So I wrote 18 deliberate sabotages of the B&O code, each one a realistic mistake a maintainer could make in good faith, and required every one to turn the suite red. Among them:

- storing the rate in basis points, both rounding directions
- taxing the ATM at the retail rate, and swapping the two rates
- **booking B&O as a liability only, the way sales tax is booked** — the exact error you were pointing at, and the nastiest of the set because it *balances perfectly* and looks tidy
- neutering the expense check so it always says "yes" — a verifier that approves everything
- expensing the tax a second time when it is paid
- removing each refusal guard in turn

Result: 17 of the 18 were caught. One escaped — and it was worth finding.

The one that escaped, and what it taught me

Mutation 10 disabled the guard that refuses **negative gross receipts**, and the test suite stayed **green**. A guard protecting against exactly the kind of corrupt figure that appears in your own history (negative ATM cash, negative inventory) could have been deleted and nothing would have complained.

The reason is subtle and worth understanding, because it is a trap I had set for myself. My test fed in gross receipts of `-100` and asserted that *something* was thrown containing the word "negative". When the negative-gross guard was removed, a **different** guard caught the input instead — the one checking that deductions do not exceed gross, because zero deductions really are greater than minus one hundred. And that guard's message happens to contain the word "negative" too.

So the test passed for the wrong reason. It was never testing the guard it believed it was testing. It would have kept passing forever with the real protection gone.

The fix is a class fix, not a patch. Every refusal in the B&O engine now carries a unique code — B0-E01 through B0-E14 — and every test pins the **code** rather than the wording. Matching on prose is matching on a coincidence; a code cannot be satisfied by accident, and rewording a message can no longer quietly disconnect a test from the thing it guards. I also added a test asserting that the negative-gross guard fires *before* the deductions guard, because that ordering turned out to be load-bearing, and a test that proves every one of the 14 codes is genuinely emitted by a real guard rather than sitting there as decoration.

The suite went from 47 tests to **54**, and the campaign was re-run against the hardened gates.

The re-run, and one more piece of honesty

Second run: M10 was **caught**. But a different mutant now reported "*pattern not found*" — my hardening had reworded the very guard that mutant was written to attack, so the sabotage no longer applied cleanly.

That is a small thing with a big principle attached. A mutation that cannot be applied has proved **nothing**, and if the harness had quietly counted it as a pass it would have been lying to me — exactly the kind of blind verifier these campaigns exist to catch. The harness is built to treat "pattern not found" as a **defect**, so it flagged itself. I repointed that mutant at the current code and ran a third time.

Final result: 18 mutants planted, 18 caught, 0 escaped. Baseline files restored and verified by checksum, suite green afterwards.

Where things stand

Squash merge	Done, pushed, tree hash proved identical
Full battery on main	464 files / 11,338 tests, all passing
CI on corrected main	Passed — deploy triggered and green
Commit author	Greenway Dev <dev@greenwaymarijuana.com> , verified at GitHub
B&O engine	47 tests, every line of your July return recomputed
Type check	0 errors
Lint	0 errors
New standing rules	20b, 114, 115

Warnings still open

Carried forward every report, per your standing request:

- **Warning #5** — the IRS's own sentence omits a count. Their text, not ours; not fixable by us.
- **Warning #8** — W-3 box 12b has no IRS instruction text to quote.
- **New, minor** — one pre-existing lint warning in the W-2 page (w2ChecksInOrder declared but never used). Proved pre-existing by stashing my changes and re-linting. I have not removed it, because deleting a symbol from a screen you rely on is not a safe thing to do inside an unrelated slice.
- **New, informational** — nothing posts to 51000 ATM Surcharge Income yet, same as B&O was. Flagged now so it is not a surprise later.

My open questions for you

1. **The B&O payable.** Standard accrual keeps both the expense and the payable. You said "expense and not a liability." I have kept both and explained why above. **Do you want it that way, or expensed directly when paid?**
2. **B&O rate changes.** The rates are settable rather than hardcoded. **Where would you like to set them** — the existing tax settings screen alongside excise and sales tax, or somewhere separate?
3. **The \$1,000,000 threshold.** Your service rate depends on prior-year receipts staying under a million. **Should the**

system warn you as you approach it, so a rate change never catches you by surprise?

4. **Which form next.** I am about to evaluate the 17 untaught federal boxes and only write lessons where they genuinely apply to you. Then the next physical form. **My recommendation is the 941**, because that is where the "not taught yet" markers actually appear — but say the word if you would rather see another one first.